

Sarvaarth Narang

Curriculum Vitae

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EDUCATION

Veermata Jijabai Technological Institute

B.Tech., Electronics and Telecommunication, Minor in Cybersecurity (CGPA: 9.66)

Mumbai, India

Jul. 2024 – Present

Indian Institute of Technology, Madras

B.S., Data Science and Applications (CGPA: 9.61)

Correspondence

May 2024 – Present

Reliance Foundation School

Grade XII, CBSE (98.2%)

Navi Mumbai, India

Jul. 2022 – Jul. 2024

North Point School

Grade X, ICSE (98.2%)

Navi Mumbai, India

Apr. 2018 – Jul. 2022

EXPERIENCE

Electronics Head

Society of Robotics and Automation, VJTI

June 2026 – Present

Mumbai, India

- Led electronics inventory management and procurement for SRA projects.
- Mentored multidisciplinary teams in electronics design and RF engineering.

Hardware Design Intern

Edith Defence Systems

May 2026 – July 2026

Mumbai, India

- Designed PCBs for drone electronics, including Battery Eliminator Circuits (BECs), ESCs, and a smoke stopper circuit.
- Contributed to BLDC motor FOC development and created embedded firmware for hardware bring-up and testing.
- Made minor firmware contributions, including adding hardware targets and UI improvements for ExpressLRS.

Second Year Core Member

Society of Robotics and Automation, VJTI

June 2025 – May 2026

Mumbai, India

- Lectured at workshops on embedded programming, control systems, computer vision, and robotics.
- Mentored a dedicated team for the GantryMate-200 project.

PROJECTS

GantryMate-200: Gantry-Based Screwing System | C, ESP-IDF

June 2025 – Present

- Designed and implemented a 3-axis automated gantry system (X, Y, Z) for high-precision fastening and loosening of approximately 280 miniature (1 mm) screws on sensitive LED panels.
- Developed ESP32-based firmware using ESP-IDF to enable coordinated multi-axis motion, homing routines, and precise positioning control.
- Integrated NEMA 17 stepper motors driven by DRV8825 and TMC2208 stepper drivers with microstepping, along with geared DC motors for toolhead actuation to achieve controlled torque.
- Designed and fabricated a custom PCB for motor driver integration, power distribution, and control signal routing.
- Achieved sub-millimeter positioning accuracy (0.05 mm) with strong emphasis on repeatability and safe operation near delicate circuitry.

Ambient Backscatter | RF, GNU Radio, Hack RF one

June 2026 – Present

- Developing battery-free ambient backscatter communication system operating in the FM broadcast band.

12 Hr Clock | KiCAD

Aug 2025 – Dec 2025

- Developed a precise timing system using crystal oscillators and frequency dividers to drive the clock's logic.
- Designed a custom PCB to integrate the logic circuit

ADSB Encoder | C++

June 2025 – July 2025

- Developed a program to encode aircraft messages in accordance with the Mode-S ADS-B standard.

- Developed an interactive chemistry simulation and game-based learning platform to improve conceptual understanding of chemistry.
- Implemented Python-based rule-driven logic where user inputs determine outcomes, with a final image displayed as feedback to simulate a viva-style evaluation experience.
- Designed and built educational mini-games that reinforce chemistry concepts through challenges, scoring, and progressive difficulty.
- Created an intuitive GUI using Tkinter enabling real-time user interaction, visualization, and gameplay.
- Implemented database architecture using both MySQL and Firebase for efficient data storage and retrieval.
- Showcased the project at the CBSE National Science Exhibition, earning recognition for innovation and educational impact.

TECHNICAL SKILLS

Languages: C, C++, Verilog, Java, Python, SQL

Embedded Systems: ESP32, ESP-IDF, STM32CubeIDE

PCB Design: kiCAD

Developer Tools: GNU Radio, LTSpice, Google Firebase, VS Code, PyCharm, STM32 MCSDK

ACHIEVEMENTS

- **National Level CBSE Science Exhibiton winner** for project CHEMSIM
- Awarded the **Dhirubhai Ambani Scholarship** for achieving the highest marks in school.
- Won the **SRA Autosim Challenge** for developing a line-following robot that completed the track in the least time.